EXT COMMUTES WITH A FLAT BASE CHANGE BELOW THE PSEUDO-COHERENCE BOUND

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

Editorial note. The citation fields (source_id, source_version, authors, source_url, license.name, source_title) are identical to those of every other card this lane has produced from this chapter, so the batch selector treats them as one source.

Allowed standard prerequisites (editorial).

- A module admits a resolution by free modules, and for a pseudo-coherent module the resolution can be chosen finitely free in a range. the author cites Algebra, Lemma algebra-lemma-resolution-by-finite-free at proof line 16681. Provides the resolution used to compute both sides.
- The Hom complex out of a free resolution computes Ext, and cohomology commutes with tensoring by a flat module. used at proof lines 16681-16695. Makes sense of the two sides and of the comparison map.
- Finitely presented modules are pseudo-coherent at the lowest bound. the author cites Lemma lemma-n-pseudo-module at the end of the proof. Reduces the first part to the second.

Source reference. Original lemma 087Q: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10435–10435. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00LP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 795–795. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 064T: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7877–7877. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let R be a ring. Let L, M, N be R-modules.*

(1) *If M is finitely presented and L is flat, then the canonical map $\mathrm{Hom}_R(M, N) \otimes_R L \to \mathrm{Hom}_R(M, N \otimes_R L)$ is an isomorphism.*
(2) *If M is $(-m)$-pseudo-coherent and L is flat, then the canonical map $\mathrm{Ext}^i_R(M, N) \otimes_R L \to \mathrm{Ext}^i_R(M, N \otimes_R L)$ is an isomorphism for $i < m$.*

Proof. Choose a resolution $F_\bullet \to M$ whose terms are free R-modules, see Algebra, Lemma 00LP. The complex $\mathrm{Hom}_R(F_\bullet, N)$ computes $\mathrm{Ext}^i_R(M, N)$ and the complex $\mathrm{Hom}_R(F_\bullet, N \otimes_R L)$ computes $\mathrm{Ext}^i_R(M, N \otimes_R L)$. There always is a map of cochain complexes

$$\mathrm{Hom}_R(F_\bullet, N) \otimes_R L \longrightarrow \mathrm{Hom}_R(F_\bullet, N \otimes_R L)$$

which induces canonical maps $\mathrm{Ext}^i_R(M, N) \otimes_R L \to \mathrm{Ext}^i_R(M, N \otimes_R L)$ for all $i \geq 0$ (canonical for example in the sense that these maps do not depend on the choice of the resolution $F_\bullet$). If L is flat, then the complex $\mathrm{Hom}_R(F_\bullet, N) \otimes_R L$ computes $\mathrm{Ext}^i_R(M, N) \otimes_R L$ since taking cohomology commutes with tensoring by L.

Having said all of the above, if M is $(-m)$-pseudo-coherent, then we may choose $F_\bullet$ such that F_i is finite free for $i = 0, \ldots, m$. Then the map of cochain complexes displayed above is an isomorphism in degrees $\leq m$ and hence an isomorphism on cohomology groups in degrees $< m$. This proves (2). If M is finitely presented, then M is (-1)-pseudo-coherent by Lemma 064T and we get the result because $\mathrm{Hom} = \mathrm{Ext}^0$. $\square$